|a_n| = |a_{n−1} + 1|

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The sequence (a_n) satisfies $a_1 = 0, |a_2| = |a_1 + 1|, \ldots, |a_n| = |a_{n-1} + 1|$. Show that

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geq -\frac{1}{2}.$$